

Project 1 R file

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```
knitr::opts_chunk$set(echo = TRUE)
```

```
library(ggplot2)
library(tidyr)
library(readxl)
library(readr)
library(dplyr)
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
## filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
## intersect, setdiff, setequal, union
```

```
setwd("/Users/chelseaoliveira/Downloads")
```

```
edu_raw <- read_csv("Education70_23.csv", skip = 3)
```

```
## Rows: 3294 Columns: 55
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): FIPS Code, State, Area name
```

```
## dbl (28): 2013 Urban Influence Code, 2024 Urban Influence Code, 2013 Rural-u...
```

```
## num (24): Less than a high school diploma, 1970, High school diploma only, 1...
```

```
##
```

```
## i Use 'spec()' to retrieve the full column specification for this data.
```

```
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
unemp_raw <- read_csv("Unemployment2023_project1.csv")
```

```
## Rows: 3286 Columns: 104
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (3): FIPS_Code, State, Area_Name
## dbl (28): Rural_Urban_Continuum_Code_2023, Urban_Influence_Code_2013, Metro...
## num (73): Civilian_labor_force_2000, Employed_2000, Unemployed_2000, Civilia...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
colnames(edu_raw)
```

```
## [1] "FIPS Code"
## [2] "State"
## [3] "Area name"
## [4] "2013 Urban Influence Code"
## [5] "2024 Urban Influence Code"
## [6] "2013 Rural-urban Continuum Code"
## [7] "2023 Rural-urban Continuum Code"
## [8] "Less than a high school diploma, 1970"
## [9] "High school diploma only, 1970"
## [10] "Some college (1-3 years), 1970"
## [11] "Four years of college or higher, 1970"
## [12] "Percent of adults with less than a high school diploma, 1970"
## [13] "Percent of adults with a high school diploma only, 1970"
## [14] "Percent of adults completing some college (1-3 years), 1970"
## [15] "Percent of adults completing four years of college or higher, 1970"
## [16] "Less than a high school diploma, 1980"
## [17] "High school diploma only, 1980"
## [18] "Some college (1-3 years), 1980"
## [19] "Four years of college or higher, 1980"
## [20] "Percent of adults with less than a high school diploma, 1980"
## [21] "Percent of adults with a high school diploma only, 1980"
## [22] "Percent of adults completing some college (1-3 years), 1980"
## [23] "Percent of adults completing four years of college or higher, 1980"
## [24] "Less than high school graduate, 1990"
## [25] "High school graduate (or equivalency), 1990"
## [26] "Some college or associate degree, 1990"
## [27] "Bachelor's degree or higher, 1990"
## [28] "Percent of adults who are not high school graduates, 1990"
## [29] "Percent of adults who are high school graduates (or equivalent), 1990"
## [30] "Percent of adults completing some college or associate degree, 1990"
## [31] "Percent of adults with a bachelor's degree or higher, 1990"
## [32] "Less than high school graduate, 2000"
## [33] "High school graduate (or equivalency), 2000"
## [34] "Some college or associate degree, 2000"
## [35] "Bachelor's degree or higher, 2000"
## [36] "Percent of adults who are not high school graduates, 2000"
## [37] "Percent of adults who are high school graduates (or equivalent), 2000"
## [38] "Percent of adults completing some college or associate degree, 2000"
## [39] "Percent of adults with a bachelor's degree or higher, 2000"
## [40] "Less than high school graduate, 2008-12"
## [41] "High school graduate (or equivalency), 2008-12"
## [42] "Some college or associate degree, 2008-12"
## [43] "Bachelor's degree or higher, 2008-12"
## [44] "Percent of adults who are not high school graduates, 2008-12"
## [45] "Percent of adults who are high school graduates (or equivalent), 2008-12"
```

```
## [46] "Percent of adults completing some college or associate degree, 2008-12"
## [47] "Percent of adults with a bachelor's degree or higher, 2008-12"
## [48] "Less than high school graduate, 2019-23"
## [49] "High school graduate (or equivalency), 2019-23"
## [50] "Some college or associate degree, 2019-23"
## [51] "Bachelor's degree or higher, 2019-23"
## [52] "Percent of adults who are not high school graduates, 2019-23"
## [53] "Percent of adults who are high school graduates (or equivalent), 2019-23"
## [54] "Percent of adults completing some college or associate degree, 2019-23"
## [55] "Percent of adults with a bachelor's degree or higher, 2019-23"
```

```
edu_clean <- edu_raw |> select(`FIPS Code`, pct_no_hs = `Percent of adults who are not high school grad
edu_clean <- edu_clean |> filter(nchar(FIPS_Code) ==5, !grepl("000$", FIPS_Code)) |> filter(!is.na(pct_
class(edu_clean$FIPS_Code)
```

```
## [1] "character"
```

```
unemp_raw <- read_csv("Unemployment2023_project1.csv")
```

```
## Rows: 3286 Columns: 104
## -- Column specification -----
## Delimiter: ","
## chr (3): FIPS_Code, State, Area_Name
## dbl (28): Rural_Urban_Continuum_Code_2023, Urban_Influence_Code_2013, Metro_...
## num (73): Civilian_labor_force_2000, Employed_2000, Unemployed_2000, Civilia...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
unemp_clean <- unemp_raw |> filter(nchar(FIPS_Code) == 5, !grepl("000$", FIPS_Code)) |> select(FIPS_Cod
class(unemp_clean$FIPS_Code)
```

```
## [1] "character"
```

```
merged_clean <- unemp_clean |> left_join(edu_clean, by = "FIPS_Code")
merged_clean <- merged_clean |> filter(!is.na(pct_full_uni), !is.na(pct_no_hs))

model <- lm(unemp_rate ~ rucc + pct_full_uni, data = merged_clean)
#summary(model)
#plot(model)

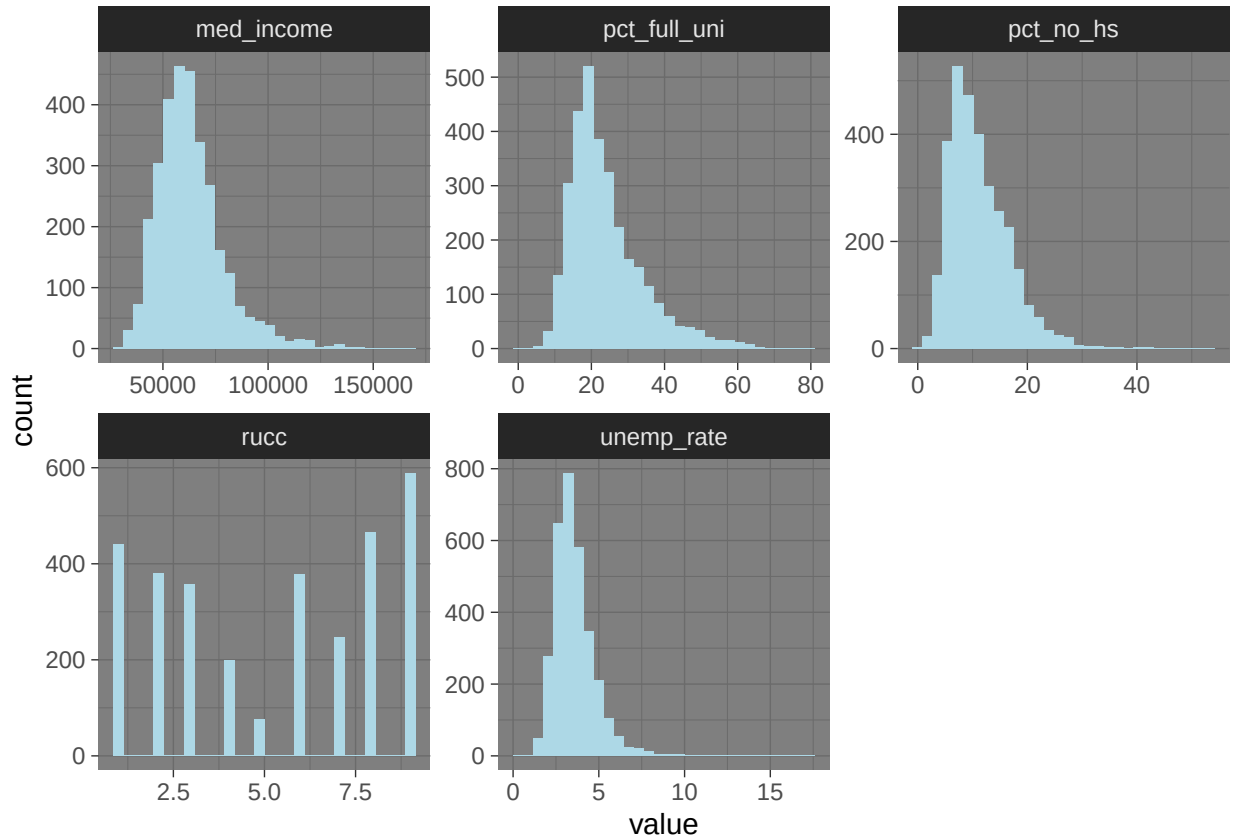
"Median_Household_Income_2022" %in% colnames(unemp_raw)
```

```
## [1] TRUE
```

```
unemp_clean <- unemp_raw |> filter(nchar(FIPS_Code) == 5, !grepl("000$", FIPS_Code)) |> select(FIPS_Cod
merged_clean <- unemp_clean |> left_join(edu_clean, by = "FIPS_Code") |> filter(!is.na(pct_full_uni), !
colnames(merged_clean)
```

```
## [1] "FIPS_Code"      "State"          "Area_Name"      "rucc"           "unemp_rate"
## [6] "med_income"     "pct_no_hs"      "pct_yes_hs"     "pct_some_uni"  "pct_full_uni"
```

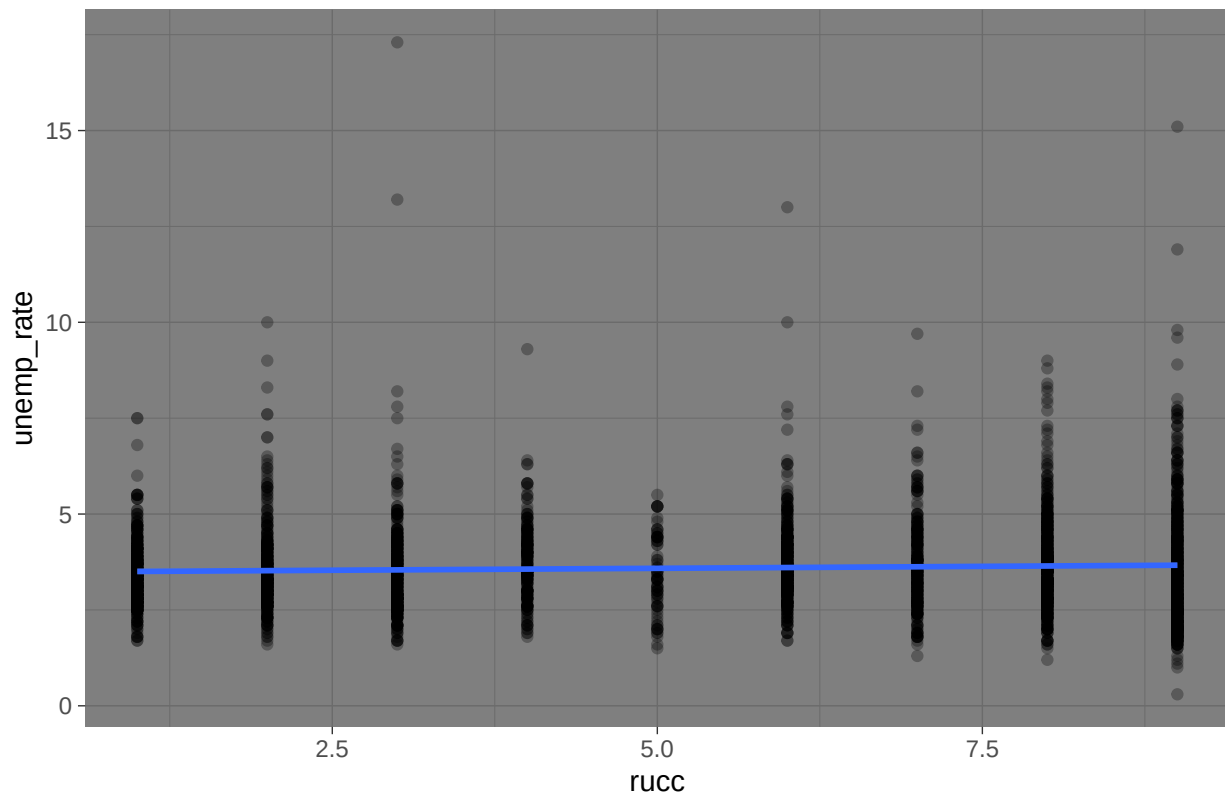
```
merged_clean |> select(unemp_rate, rucc, pct_full_uni, pct_no_hs, med_income) |> gather() |> ggplot(aes
```



```
ggplot(merged_clean, aes(x= rucc, y = unemp_rate))) + geom_point(alpha = 0.3) + geom_smooth(method = "lm
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

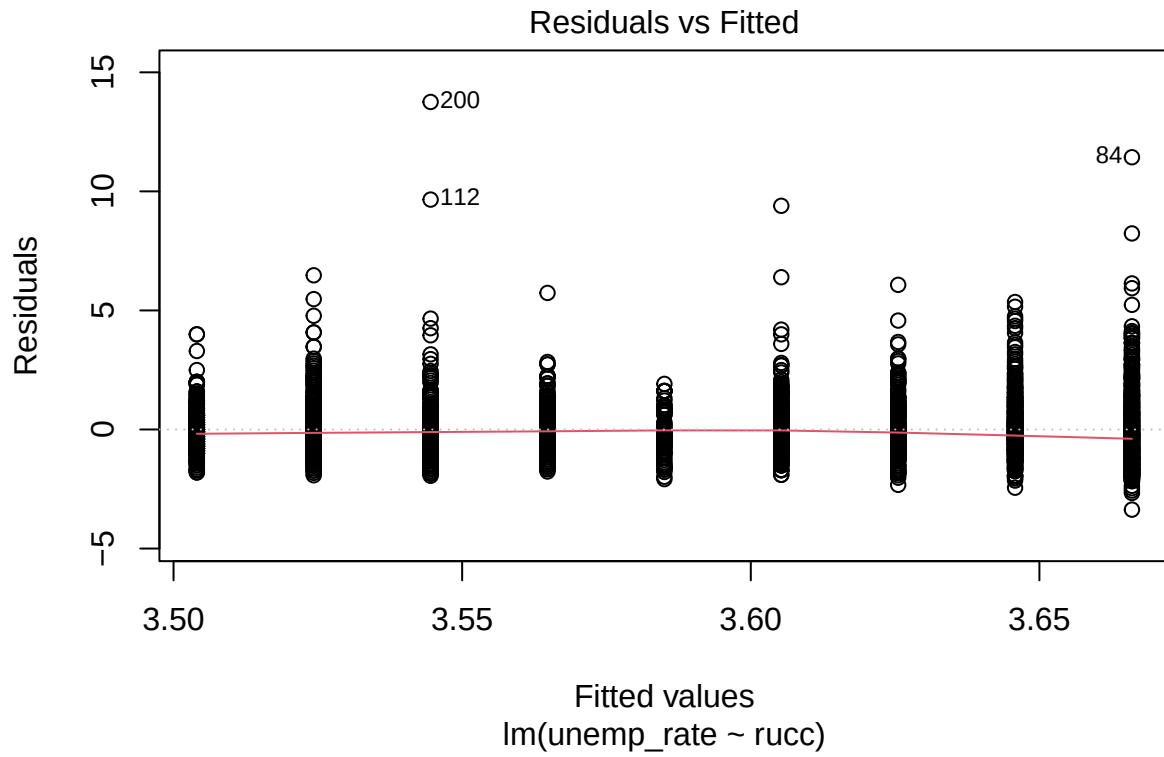
Unemployment and Rurality

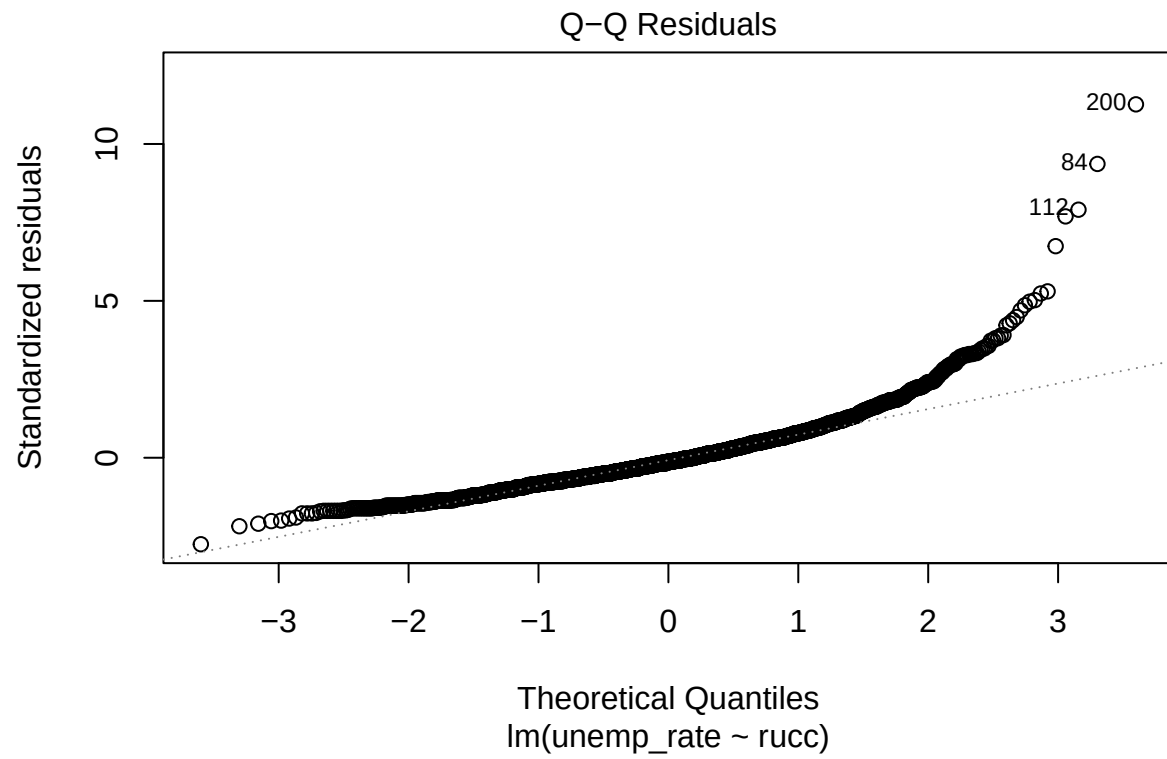


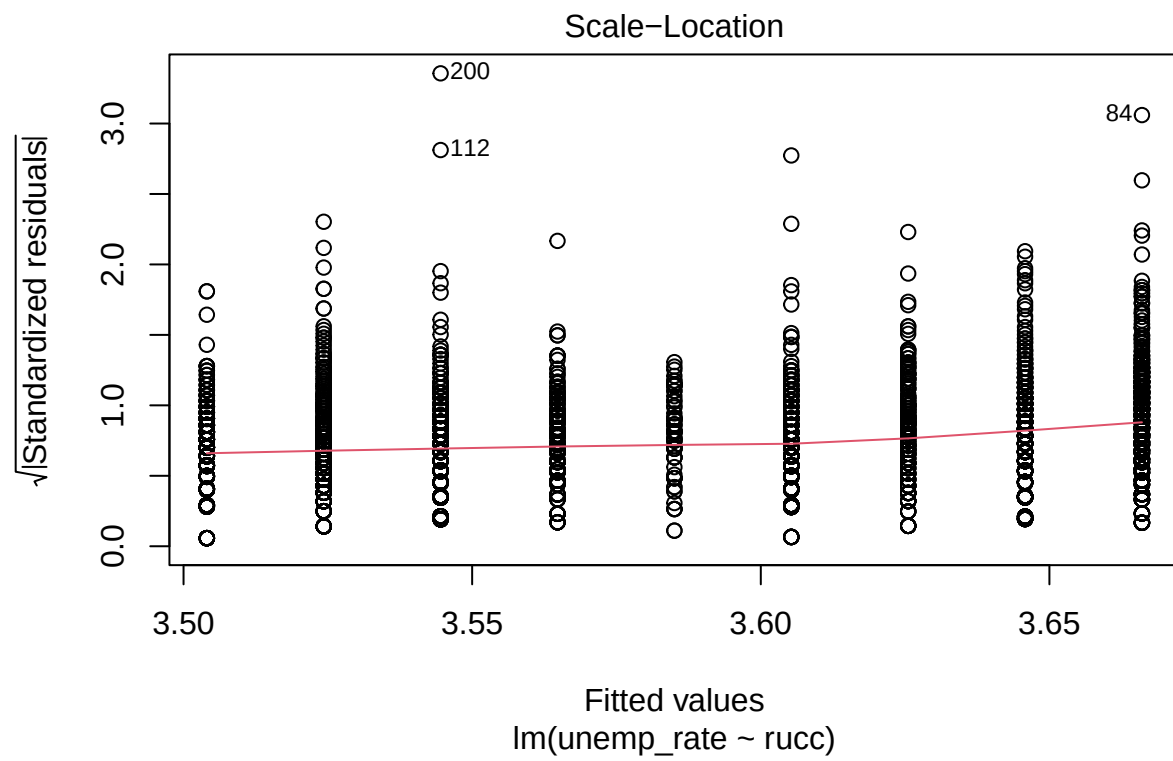
```
model11 <- lm(unemp_rate ~ rucc, data = merged_clean)
summary(model11)
```

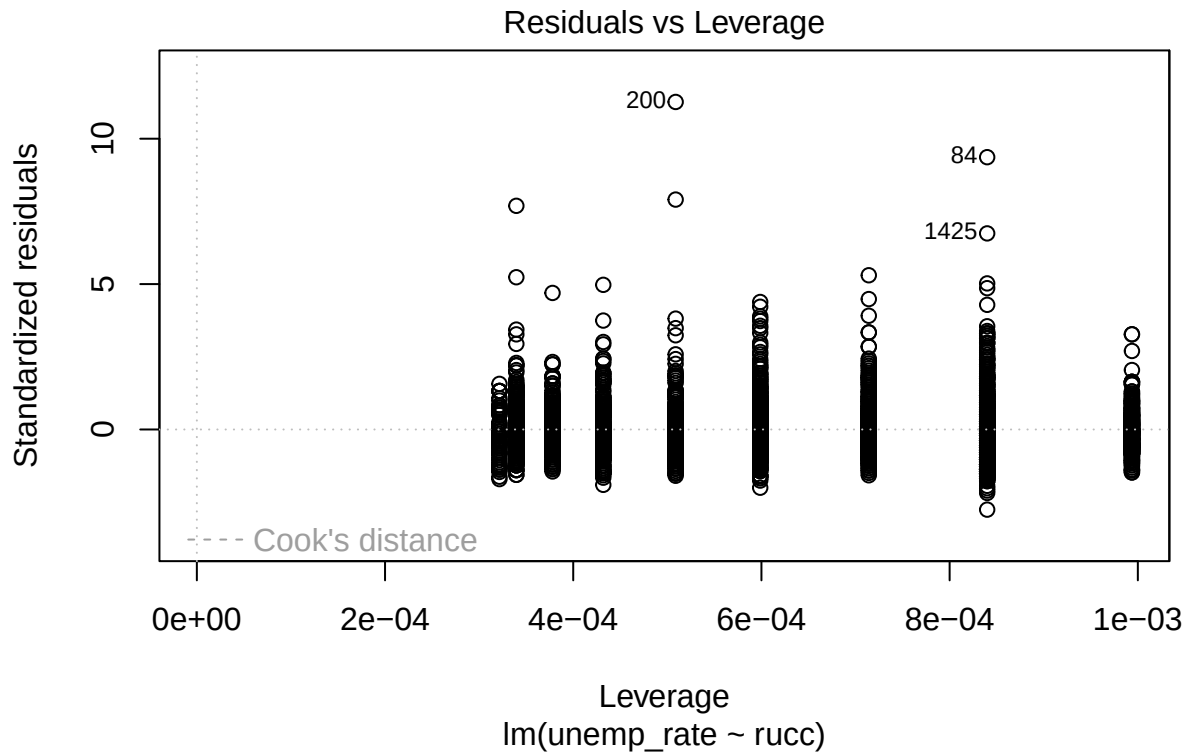
```
##
## Call:
## lm(formula = unemp_rate ~ rucc, data = merged_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.3660 -0.7660 -0.1654  0.5757 13.7555
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.483786   0.044846  77.683  < 2e-16 ***
## rucc         0.020250   0.007451   2.718  0.00661 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.222 on 3132 degrees of freedom
## Multiple R-squared:  0.002353,    Adjusted R-squared:  0.002034
## F-statistic: 7.386 on 1 and 3132 DF,  p-value: 0.006609
```

```
plot(model11)
```





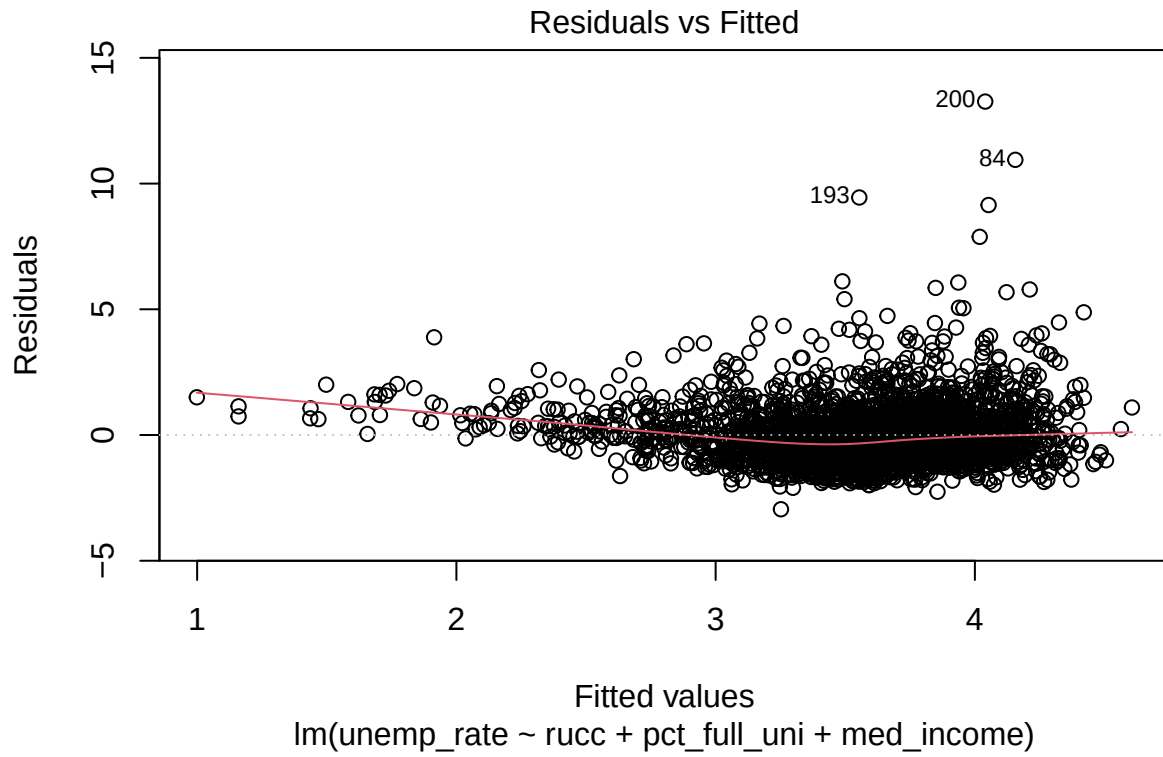


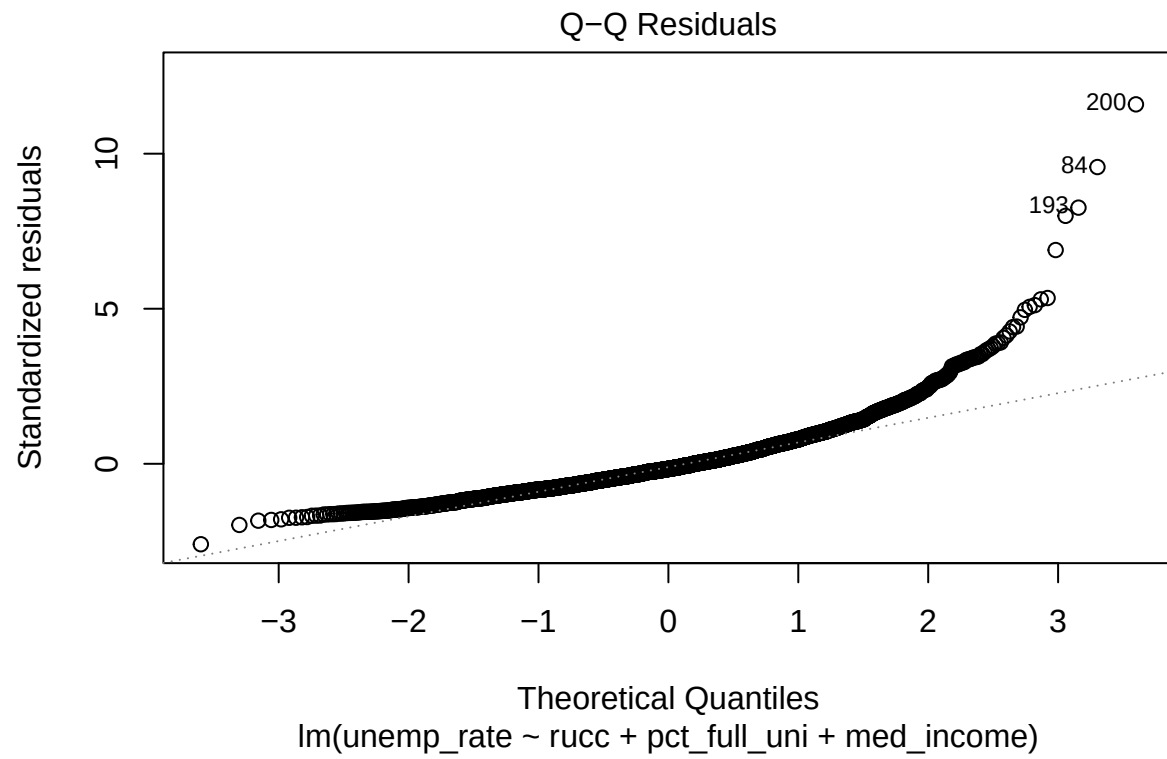


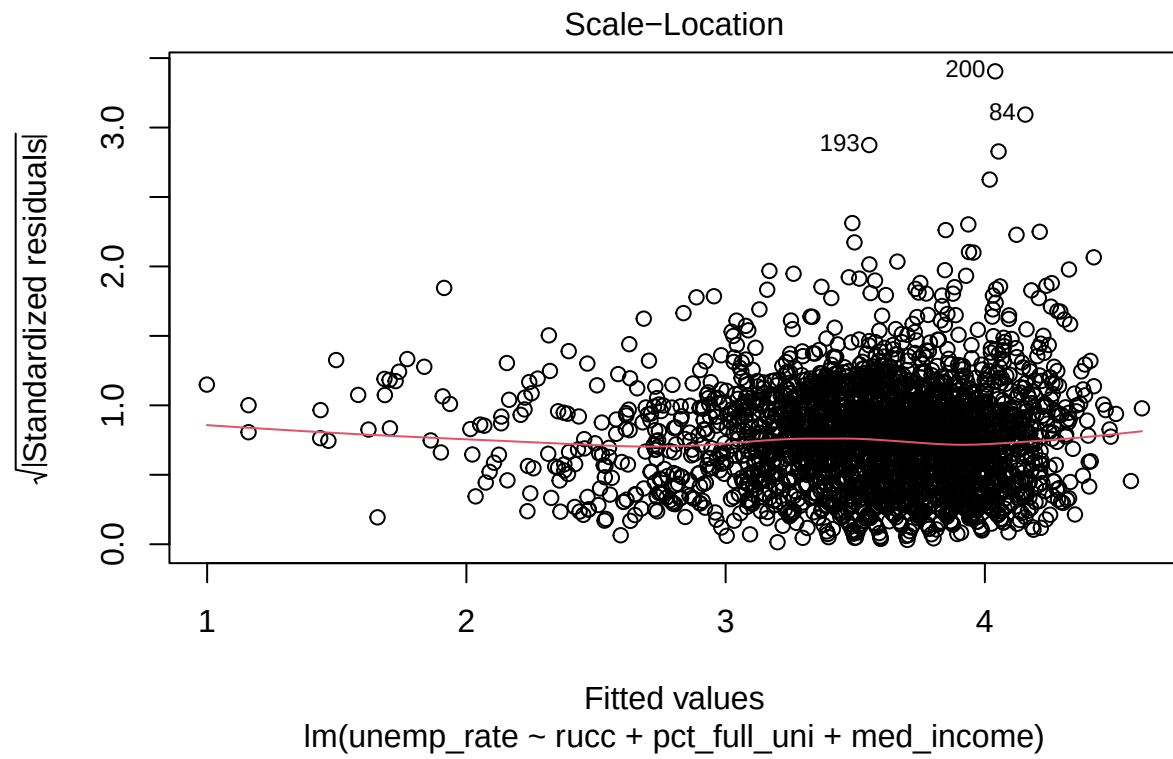
```
model2 <- lm(unemp_rate ~ rucc + pct_full_uni + med_income, data = merged_clean)
summary(model2)
```

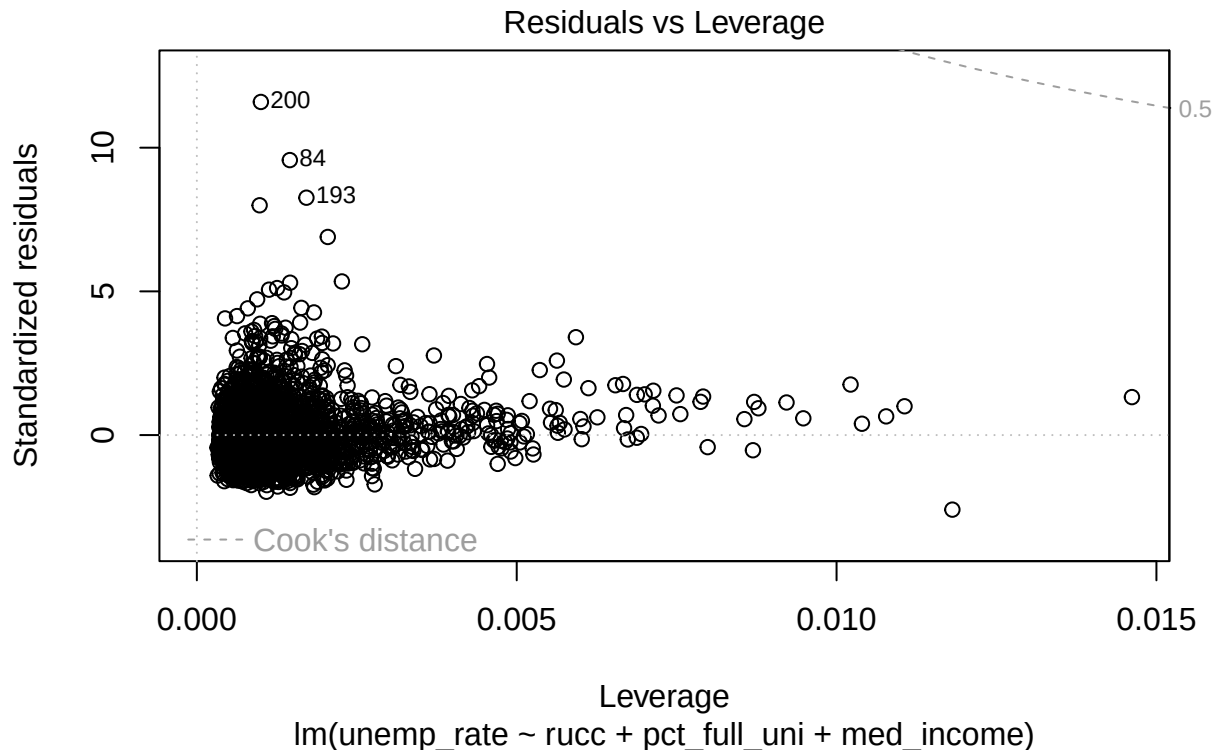
```
##
## Call:
## lm(formula = unemp_rate ~ rucc + pct_full_uni + med_income, data = merged_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9520 -0.7355 -0.1882  0.4912 13.2602
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.629e+00  1.169e-01  48.154 < 2e-16 ***
## rucc         -6.037e-02  7.975e-03  -7.570 4.89e-14 ***
## pct_full_uni -2.070e-02  2.937e-03  -7.048 2.22e-12 ***
## med_income   -1.936e-05  1.875e-06 -10.329 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.145 on 3130 degrees of freedom
## Multiple R-squared:  0.1247, Adjusted R-squared:  0.1239
## F-statistic: 148.7 on 3 and 3130 DF, p-value: < 2.2e-16
```

```
plot(model2)
```









```
merged_clean$log_unemp_rate <- log(merged_clean$unemp_rate)
model3 <- lm(log_unemp_rate ~ rucc + pct_full_uni + med_income, data=merged_clean)
summary(model3)
```

```
##
## Call:
## lm(formula = log_unemp_rate ~ rucc + pct_full_uni + med_income,
##     data = merged_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.27329 -0.18324 -0.01183  0.17051  1.48359
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.858e+00  2.956e-02  62.872 < 2e-16 ***
## rucc         -2.303e-02  2.016e-03 -11.420 < 2e-16 ***
## pct_full_uni -5.037e-03  7.426e-04  -6.784 1.4e-11 ***
## med_income   -6.142e-06  4.740e-07 -12.959 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2894 on 3130 degrees of freedom
## Multiple R-squared:  0.1546, Adjusted R-squared:  0.1538
## F-statistic: 190.7 on 3 and 3130 DF, p-value: < 2.2e-16
```

```
plot(model13)
```

